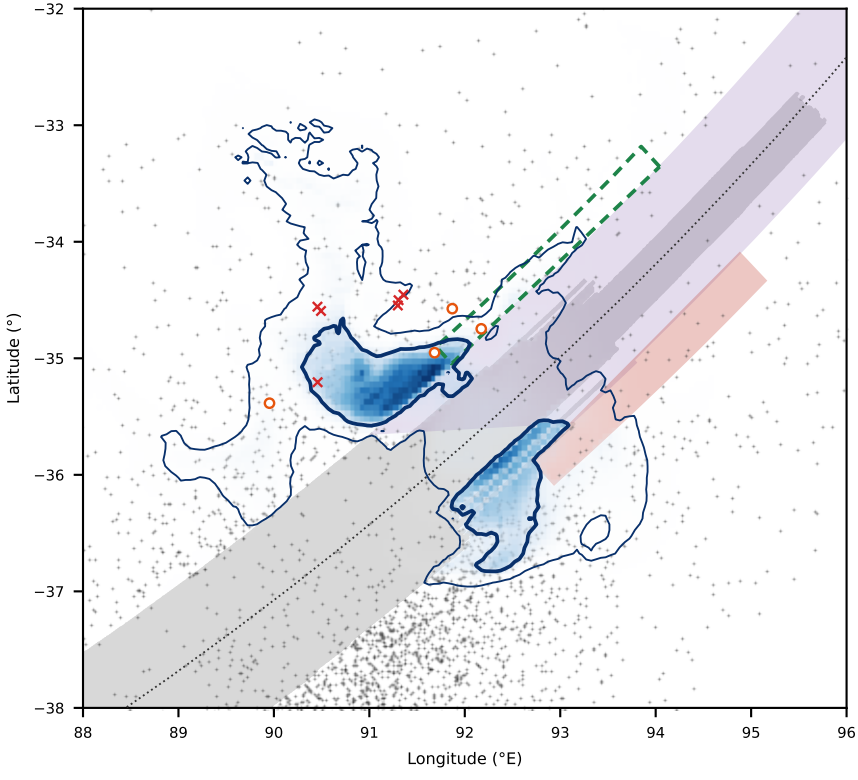
If the imaged objects are from 9M-MRO: where the debris entered the sea — 00:19 Held Out, impacts faster than 212 m/s removed (no structural limit in the model)
Pléiades objects + all four COSMO-SkyMed contacts, one debris field
after Phase 2 + Bluefin-21 + OI 2018 + 2025-26 (grade C)



strata weights	re-weighted (drawn)	fixed
50 % region	15,759 km²	15,759 km²
90 % region	82,792 km²	82,792 km²
mean	35.37 S 91.62 E	35.37 S 91.62 E
outside past searches	81.2 %	81.2 %
in OI north-west band	2.9 %	2.9 %
searches leave, under H	0.496	0.496
tension, ln S (p)	-0.58 (p 0.28)	-0.58 (p 0.28)
mean shift from flight PDF	81 NM	81 NM

probability density under H (relative)
50 % region under H
90 % region under H
impact PDF without H (sampled points)
ATSB Phase 2 2014-17 (official)
Ocean Infinity 2018 (grade C)
OI 2025-26 south-east band (grade C)
OI north-west band (not searched; grade C)
7th arc, FL400
Pléiades rating-5 objects
COSMO-SkyMed radar contacts (all four)

Source: impacts next-run-c (strata next-c-free, next-c-repro-radar, next-c-descent-climb, next-c-routes; provenance read from next-c-free) (8 seeds pooled with equal weight; source run runs/C8-next-c-free, code 097a5d77); prior track 289.7 deg (sd 1.0); config chain (run.json): davey2016-inmarsat + no-exhaustion-prior + s1-factor + s2-temperature + s3-internal + s4-ceiling + s5-hard-reject + s6-tanks + reference-snapshots + radar-full + s7-single-engine + s8-hold-taper + inop-flow-fix + free + server-c-free + hypotheses/end-of-flight/smoke/snapshot-m0011.toml + hypotheses/end-of-flight/smoke/terminal.toml + hypotheses/end-of-flight/full/reference-289.toml + hypotheses/end-of-flight/full/descent-idle-floor.toml + hypotheses/end-of-flight/full/seed-1.toml + hypotheses/end-of-flight/smoke/v2-broad.toml + hypotheses/end-of-flight/full/family-b-ditching.toml + hypotheses/end-of-flight/full/residual-bank-boeing.toml; tempered sampler (m1839, m1941, m2041, m2141, m2241, m0011, 16 stages); particles per mode [1000000, 500000, 2500000, 500000, 2500000]; BFO bias (run.json bfo_bias): {"mean_hz":150.0,"sd_hz":25.0}.

What was run: 00:19 Held Out: none of the 00:19 BTO/BFO values scored (held out); log-on not from fuel exhaustion (no lag term); aircraft transmitting at 00:19:37 and not powered at 01:15:56 (the two observed facts; end of flight's 'unpowered' existence constraint, ruling ~19:10 UTC B (b)); impacts faster than 212 m/s removed (no structural limit in the model), the rest re-weighted (ruling 1(C), 1.25 VMO) [end-of-flight arm: none+unpowered]. Strata mixed by P(family) re-weighted by the 00:19 evidence (ruling C; end of flight's family-evidence-run-c.json, key '00:19 Held Out +unpowered ~v212': next-c-free 0.5891, next-c-repro-radar 0.2523, next-c-descent-climb 0.1347, next-c-routes 0.0240); the core's fixed P(family) (next-c-free 0.5890, next-c-repro-radar 0.2520, next-c-descent-climb 0.1350, next-c-routes 0.0240) is shown beside it. The search then re-weights strata by their own evidence.

Labels: run C: core 66d77b5 option C (free 7M x 8 seeds, others 3.5M x 8), split-half NOT converged (floor 0.924: free 0.918, Davey dynamics + radar 0.910, descent-climb 0.892, routes 0.850); internal-v1 fuel with one-engine flow correction; EoF run C recipe (PROVISIONAL physics; descents above 6,500 ft/min and unloading not reachable); wide Pléiades grid 78-115 E, 45-5 S.

Search: the searched-areas module's own per-impact likelihood (point target, rho = 0.05, shared). Phase 2 (q 0.945), Bluefin-21 (q 0.9) official; OI 2018 (q 0.9, coverage 0.889) and OI 2025-26 south-east band (q 0.9, coverage 0.7808) are community tracings, grade C; the north-west band is drawn only, never negative evidence.

Under H: 12 rating-5 Pléiades objects in 6 clusters (3 km), equal weights; COSMO-SkyMed contacts equally weighted, dawn 20 / dusk 21 Mar passes equal (unverified source, time, footprint); 'one debris field' = likelihoods multiplied per ocean model; GLORYS12 + ERA5 and GlobCurrent daily + ERA5 equal weight; measured transport error (~95-120 km per component at 15 d); windage 0-5 %; Pléiades and COSMO errors independent (rho = 0).

Conditional on H (the objects are debris from 9M-MRO); no Bayes factor or P(H | data).

Background: the impact PDF without H (flight posterior alone, after the same searches), as 6,000 points drawn in proportion to its probability from 0.05 deg grid cells and jittered within each cell. 39 % of it lies inside this frame.